ASCO rapid recommendations

# Chemotherapy and Targeted Therapy for Human Epidermal Growth Factor Receptor 2–Negative Metastatic Breast Cancer That Is Either Endocrine-Pretreated or Hormone Receptor–Negative: ASCO Guideline Rapid Recommendation Update

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*ASCO Rapid Recommendations Updates highlight revisions to select ASCO guideline recommendations as a response to the emergence of new and practice-changing data. The rapid updates are supported by an evidence review and follow the guideline development processes outlined in the ASCO Guideline Methodology Manual. The goal of these articles is to disseminate updated recommendations, in a timely manner, to better inform health practitioners and the public on the best available cancer care options.*

**ASSOCIATED CONTENT**

**The companion to this article was published in the December 10, 2021 issue of *Journal of Clinical Oncology*. See accompanying article on page 3938**

**Appendix**

Author affiliations and support information (if applicable) appear at the end of this article.

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## BACKGROUND

In 2021, ASCO published a guideline on chemotherapy and targeted therapy for patients with human epidermal growth factor receptor 2 (HER2)–negative metastatic breast cancer that is either endocrine-pretreated or hormone receptor–negative.[1] The DESTINY-Breast04 was a phase III, two-group, open-label, randomized multicenter trial that compared trastuzumab deruxtecan with the treatment of physician's choice (TPC, limited to capecitabine, eribulin, gemcitabine, paclitaxel, or nab-paclitaxel), in patients with metastatic HER2-low breast cancer (defined as HER2 immunohistochemistry [IHC] 1+ or 2+ and in situ hybridization [ISH]–negative) who had been previously treated with one or two lines of chemotherapy in the metastatic setting. Of note, patients with metastatic hormone receptor–positive metastatic breast cancer were required to have endocrine-refractory disease. The DESTINY-Breast04 results provided a strong signal indicating the need to update the 2021 guideline recommendations.[2]

## METHODS

A targeted electronic literature search was conducted to identify any additional phase III randomized controlled trials of treatment options in this patient population. No additional randomized controlled trials were identified. The original guideline Expert Panel was reconvened to review new evidence from DESTINY-Breast04 and to review and approve the revised recommendation.

## EVIDENCE REVIEW

In this trial, 494 patients with hormone receptor–positive and 63 patients with hormone receptor–negative metastatic breast cancer and without evidence in the primary tumor or a metastatic recurrence of HER2 overexpression by IHC (if IHC 1+) or by ISH (if IHC 2+ with a negative ISH reflex test) were randomly assigned 2:1 to either trastuzumab deruxtecan or TPC as defined above. Patients whose tumors tested HER2 IHC 0 were not eligible for this trial. The primary outcome was progression-free survival (PFS) as evaluated by blinded independent review in the hormone receptor–positive cohort. Other outcomes included PFS in the combined hormone receptor–positive and –negative cohort, overall survival (OS), objective response rate, duration of response, and safety.

At a median follow-up of 18.4 months, the median treatment duration with trastuzumab deruxtecan was 8.2 months (range, 0.2-33.3) and 3.5 months (range, 0.3-17.6) with TPC.

The median PFS for hormone receptor-positive patients was 10.1 months (95% CI, 9.5 to 11.5) with trastuzumab deruxtecan and 5.4 months (95% CI, 4.4 to 7.1) for TPC (hazard ratio [HR], 0.51; 95% CI, 0.40 to 0.64; $P < .0001$). The median OS with trastuzumab deruxtecan was 23.9 months (95% CI, 20.8 to 24.8) and 17.5 months (95% CI, 15.2 to 22.4) with TPC (HR, 0.64; 95% CI, 0.48 to 0.86; $P = .0028$).

The results for the full analysis set, which included both hormone receptor–positive and hormone receptor–negative patients, reported a median PFS of 9.9 months

(95% CI, 9.0 to 11.3) with trastuzumab deruxtecan and 5.1 months (95% CI, 4.2 to 6.8) with TPC (HR, 0.50; 95% CI, 0.40 to 0.63; $P < .0001$). The median OS for the full analysis set was 23.4 months (95% CI, 20.0 to 24.8) with trastuzumab deruxtecan and 16.8 months (95% CI, 14.5 to 20.0) with TPC (HR, 0.64; 95% CI, 0.49 to 0.84; $P = .0010$). There were only 58 patients with hormone receptor–negative tumors but had broadly similar results. The median PFS for hormone receptor–negative patients was 8.5 months (95% CI, 4.3 to 11.7) with trastuzumab deruxtecan and 2.9 months (95% CI, 1.4 to 5.1) for TPC (HR, 0.46; 95% CI, 0.24 to 0.89; $P = .0135$). The median OS with trastuzumab deruxtecan was 18.2 months (95% CI, 13.6 to not estimable) and 8.3 months (95% CI, 5.6 to 20.6) with TPC (HR, 0.48; 95% CI, 0.24 to 0.95; $P = .1732$). Drug-related interstitial lung disease or pneumonitis was confirmed in 12.1% of the patients who received trastuzumab deruxtecan, including 13 patients (3.5%) with asymptomatic grade 1 events, 24 patients (6.5%) with symptomatic grade 2 events, five patients (1.3%) with severe symptoms requiring oxygen, and three patients (0.8%) who died from it.

Appraisal of the trial report using the GRADE[3] instrument found a moderate certainty of the evidence because of the open trial design and the results being based on an interim analysis only.

DESTINY-Breast04 represents the first randomized phase III trial of a HER2-directed treatment in patients with HER2 IHC 1+ or 2+ and ISH-negative metastatic breast cancer to demonstrate statistically significant PFS and OS benefits when compared with TPC independent of hormone receptor status.

## UPDATED RECOMMENDATION

Patients with HER2 IHC 1+ or 2+ and ISH-negative metastatic breast cancer who have received at least one prior chemotherapy for metastatic disease, and if hormone receptor–positive are refractory to endocrine therapy, should be offered treatment with trastuzumab deruxtecan (Type: evidence based, benefits outweigh harms; Evidence quality: moderate; Strength of recommendation: strong).

### Guideline Disclaimer

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### Guideline and Conflicts of Interest

The Expert Panel was assembled in accordance with ASCO's Conflict of Interest Policy Implementation for Clinical Practice Guidelines (Policy, found at https://www.asco.org/guideline-methodology). All members of the Expert Panel completed ASCO's disclosure form, which requires disclosure of financial and other interests, including relationships with commercial entities that are reasonably likely to experience direct regulatory or commercial impact as a result of promulgation of the guideline. Categories for disclosure include employment; leadership; stock or other ownership; honoraria, consulting, or advisory role; speaker's bureau; research funding; patents, royalties, or other intellectual property; expert testimony; travel, accommodations, and expenses; and other relationships. In accordance with the Policy, the majority of the members of the Expert Panel did not disclose any relationships constituting a conflict under the Policy.

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## EDITOR'S NOTE

This ASCO Clinical Practice Guideline Recommendation Update provides a recommendation update, with review and analysis of the relevant literature for the recommendation. Additional information, including links to patient information at www.cancer.net, is available at http://www.asco.org/breast-cancer-guidelines.

## EQUAL CONTRIBUTION

B.M. and L.A.C. were expert panel co-chairs.

## AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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## AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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## APPENDIX 1. CHEMOTHERAPY AND TARGETED THERAPY FOR PATIENTS WITH HUMAN EPIDERMAL GROWTH FACTOR RECEPTOR 2–NEGATIVE METASTATIC BREAST CANCER THAT IS EITHER ENDOCRINE-PRETREATED OR HORMONE RECEPTOR–NEGATIVE EXPERT PANEL

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